1 Vector Reduction and Un-Leveled FHE Schemes

The underlying concept is relatively straightforward and partially functional; however, a primary challenge remains regarding magnitude control. The goal is to compute homomorphic operations directly as a vector modulo another vector.

The inherent difficulty in this approach is the continuous growth (or complexity explosion) of the vector during computation. The critical research question is whether this vector can be consistently reduced to ensure its magnitude never exceeds a predefined threshold. Bounding the vector's growth is a strict prerequisite for enabling an unbounded number of operations, which is necessary to transition the construction into a fully un-leveled (Fully Homomorphic Encryption) scheme.

A major constraint complicates this process: The underlying secret modulus N must never be directly reduced, nor can its value be leaked to the evaluator.

2 The Normalization Problem via ILP

Mathematically, there is guaranteed to be a solution to the corresponding Integer Linear Programming (ILP) problem that bounds the vector's magnitude appropriately. The remaining open questions regarding this reduction step are:

1. Can public correction vectors be provided to the evaluator without leaking the secret modulus N?
2. Can a solution to this specific ILP problem be found efficiently during homomorphic evaluation?

Solving the normalization (reduction) of ciphertexts securely and efficiently is currently the most pressing cryptographic challenge in this construction.

3 Zero-Testing via the Discrete Logarithm (DLog)

An additional theoretical requirement is the implementation of a zero-test mechanism. A promising approach to achieve this relies on the Discrete Logarithm (DLog) problem.

Instead of utilizing a prime modulus, the scheme employs a composite modulus to embed a checksum into the message space. Let the structure be denoted as $M = [c, m]$. During the evaluation phase, an evaluation key is utilized to simultaneously evaluate M alongside a randomized component. The evaluation function takes the form:

$$\text{eval}(c) = k_1 \cdot [c, m] + k_2 \cdot [r, r] \tag{1}$$

Subsequently, one computes an element g such that its order evaluates to:

$$\text{ord}(g) = [0, m] \tag{2}$$

While the exact details of the zero-test can likely be formalized using this or similar group-theoretic constructions, achieving efficient ciphertext normalization (as outlined in Section 2) remains the primary focus for proving the viability of the scheme.